

AI-Based Smart Wardrobe and Outfit Recommendation System

AGENDA

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- Methodology
- System Architecture
- Data Flow Diagrams
- Implementation
- Results
- Testing and Validation
- Conclusion
- Future work
- Acknowledgement
- References

PROBLEM STATEMENT

- In today's fast-paced lifestyle, individuals often face difficulty in selecting appropriate outfits from their existing wardrobe for different occasions.
- This challenge is influenced by factors such as poor wardrobe organization, limited fashion knowledge, uncertainty in color coordination, and difficulty in matching clothing with personal body shape and style preferences.
- Existing fashion platforms primarily focus on product browsing and purchasing, with limited support for personalized wardrobe management, intelligent outfit recommendations, or virtual visualization.
- As a result, users spend excessive time deciding what to wear, underutilize their clothing collection, and experience dissatisfaction with their fashion choices.

ABSTRACT

- Increasing demand for personalized solutions in fashion technology.
- Users face challenges in wardrobe management, outfit selection, and style identification.
- Existing platforms lack personalized styling and intelligent wardrobe support.
- Proposed system: AI-powered personal styling and wardrobe management solution.
- Enables digital wardrobe organization and context-aware outfit recommendations.
- Features include outfit generation, virtual try-on, wishlist management, and AI-based styling assistance.
- Recommendations are based on user preferences, wardrobe data, body characteristics, and color compatibility.
- Aims to improve decision-making, efficiency, and overall personalization in fashion.

LITERATURE REVIEW

Title	Authors	Description
Visually-Aware Fashion Recommendation and Design with Generative Image Models	Wang-Cheng Kang, Chen Fang, Zhaowen Wang, Julian J. McAuley	Focuses on personalized fashion recommendation using visual clothing features and generative models.
Toward Characteristic-Preserving Image-based Virtual Try-On Network	Bochao Wang, Huabin Zheng, Xiaodan Liang, Yimin Chen, Liang Lin, Meng Yang	Explains virtual try-on using AI, ensuring clothing details like texture, shape, and design are preserved while fitting outfits onto user images.
Personalized Outfit Recommendation With Learnable Anchors	Zhi Lu, Yang Hu, Yan Chen, Bing Zeng	Focuses on personalized outfit recommendations by considering user preferences and clothing compatibility.
Learning Type-Aware Embeddings for Fashion Compatibility	Mariya I. Vasileva, Bryan A. Plummer, Krishna Dusad, Shreya Rajpal, Ranjitha Kumar, David Forsyth	Studies compatibility between clothing categories like tops, bottoms, and shoes. Useful for recommending matching outfits from a wardrobe.

METHODOLOGY

Architecture Overview

- Full-stack system integrating frontend, backend, and AI services
- Backend handles data, users, and AI integration
- Frontend provides interactive wardrobe and styling interface
- AI supports image analysis, recommendations, and virtual try-on

User Authentication and Profiling

- Secure token-based authentication
- Stores personal, style, and wardrobe data
- Enables personalized recommendations

Digital Wardrobe Management

- Upload and manage clothing items
- Includes image data and metadata
- Tagging for filtering by mood, weather, and events

METHODOLOGY

AI-Based Clothing Analysis

- Analyzes clothing images
- Extracts category, color, and style
- Improves recommendations using datasets

Recommendation Engine

- Generates outfits based on compatibility and preferences
- Considers body type and color suitability

Virtual Try-On

- Visualizes outfits on user images
- Maintains realistic fit and alignment

AI Stylist

- Chatbot for styling suggestions and tips
- Adapts based on user interaction

METHODOLOGY

Outfit Generation

- Creates outfit combinations
- Filters by occasion and preferences
- Allows saving outfits

Moodboard Module

- Stores inspirations
- Supports reference-based styling

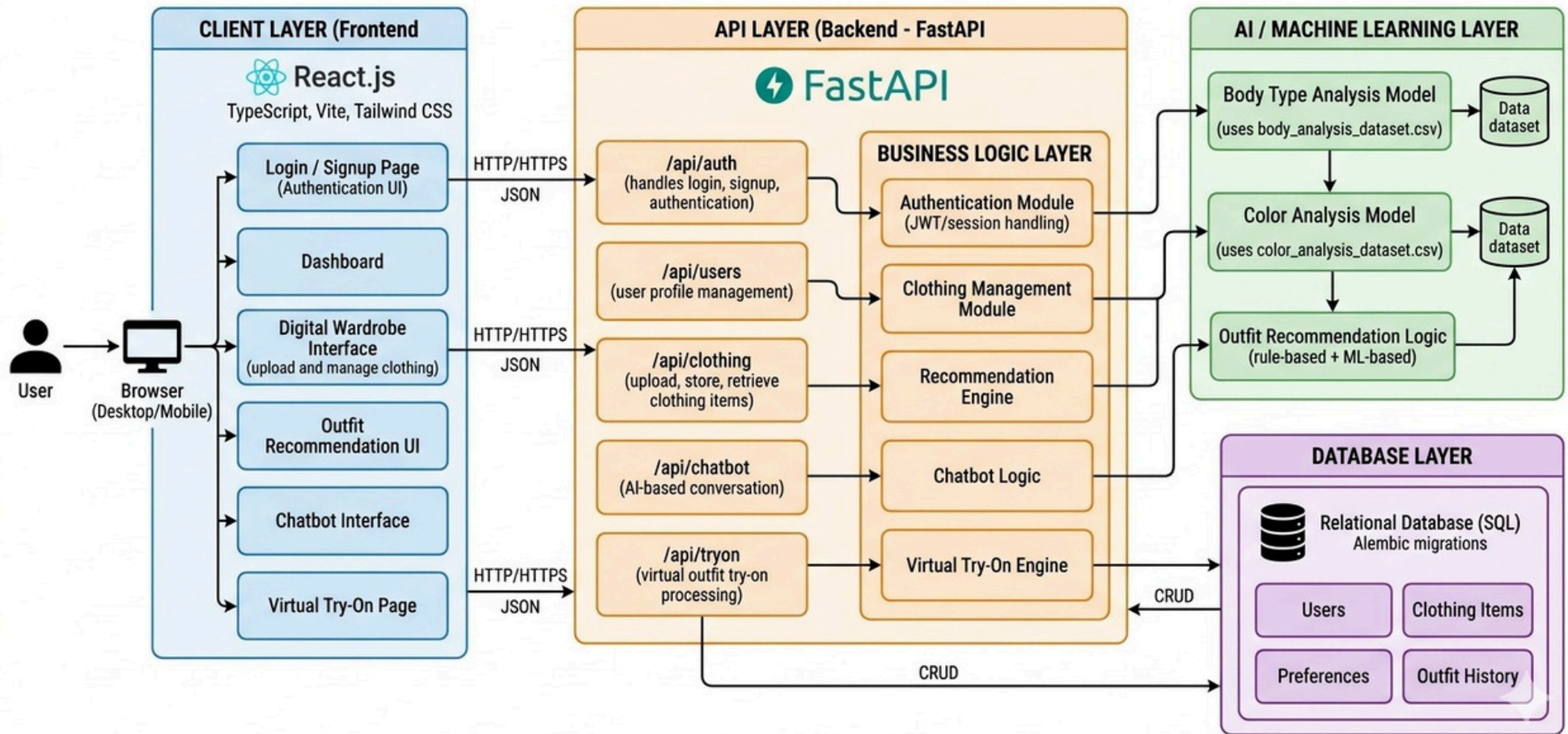
AI Intelligence Layers

- Appearance Analysis (body & color)
- Image Understanding (feature extraction)
- Compatibility Modeling (outfit matching)
- Personalization (user-based adaptation)
- Visualization (virtual try-on)

SYSTEM ARCHITECTURE

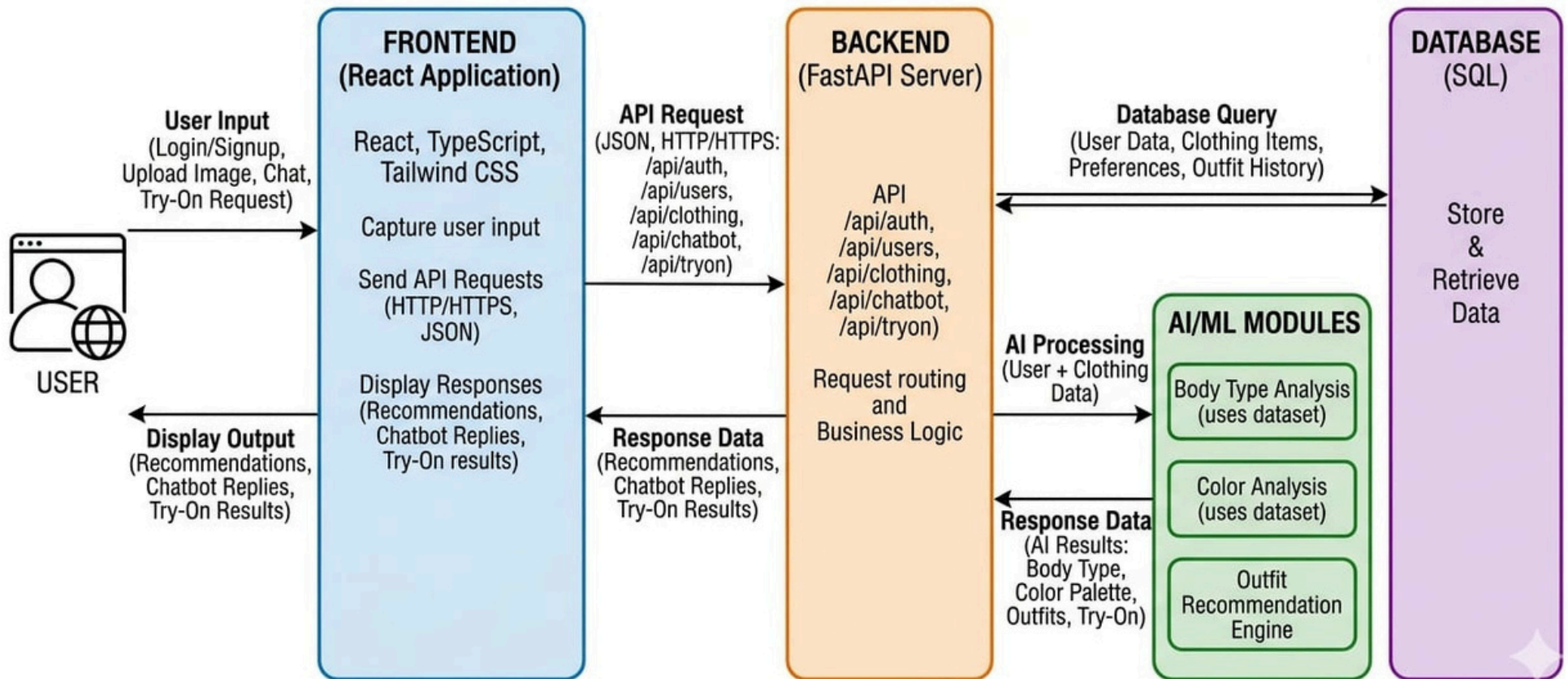
Clueless – AI Fashion Stylist

Full-stack web application



DATA FLOW DIAGRAM

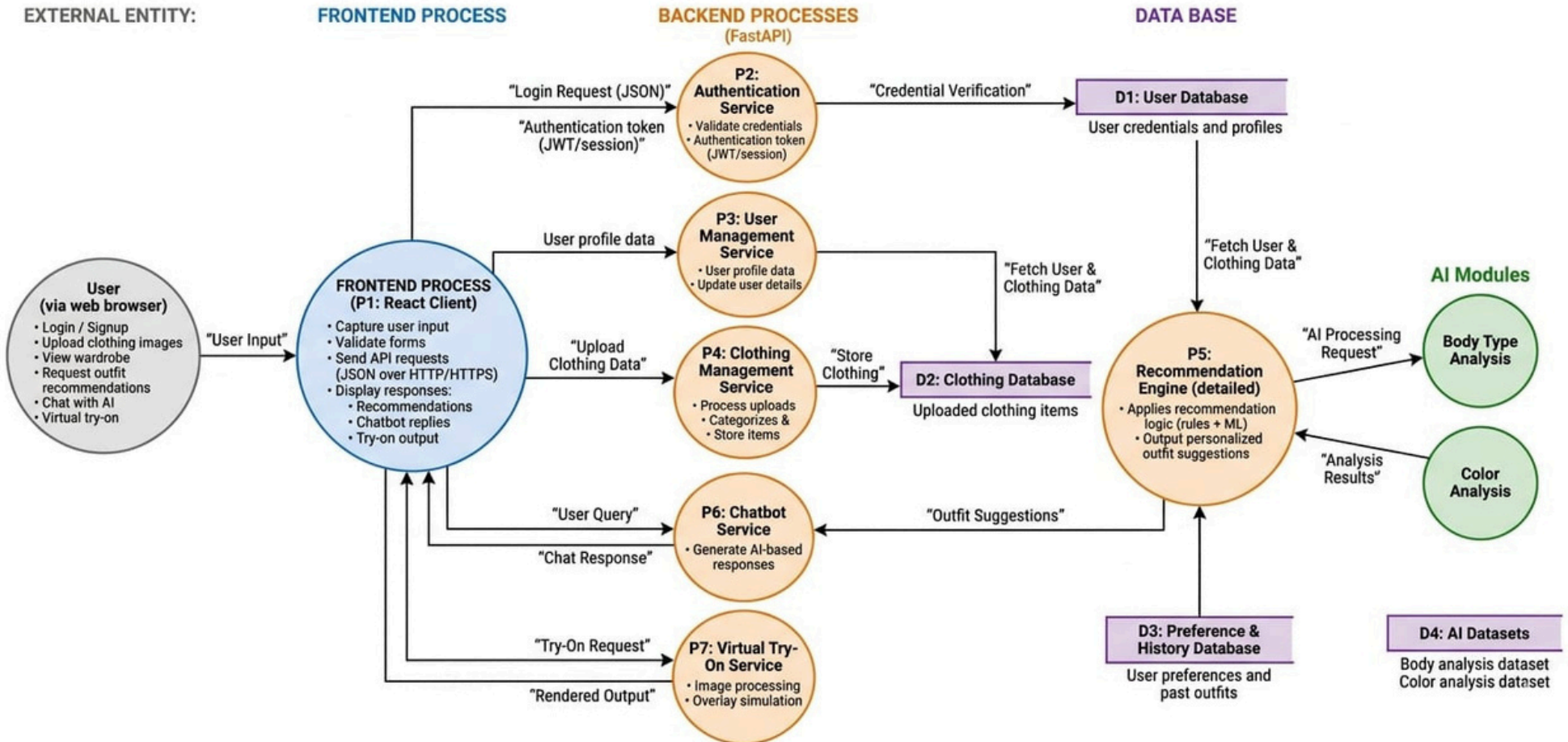
Clueless – AI Fashion Stylist Data Flow (DFD Level 1)



DATA FLOW DIAGRAM

Clueless – AI Fashion Stylist

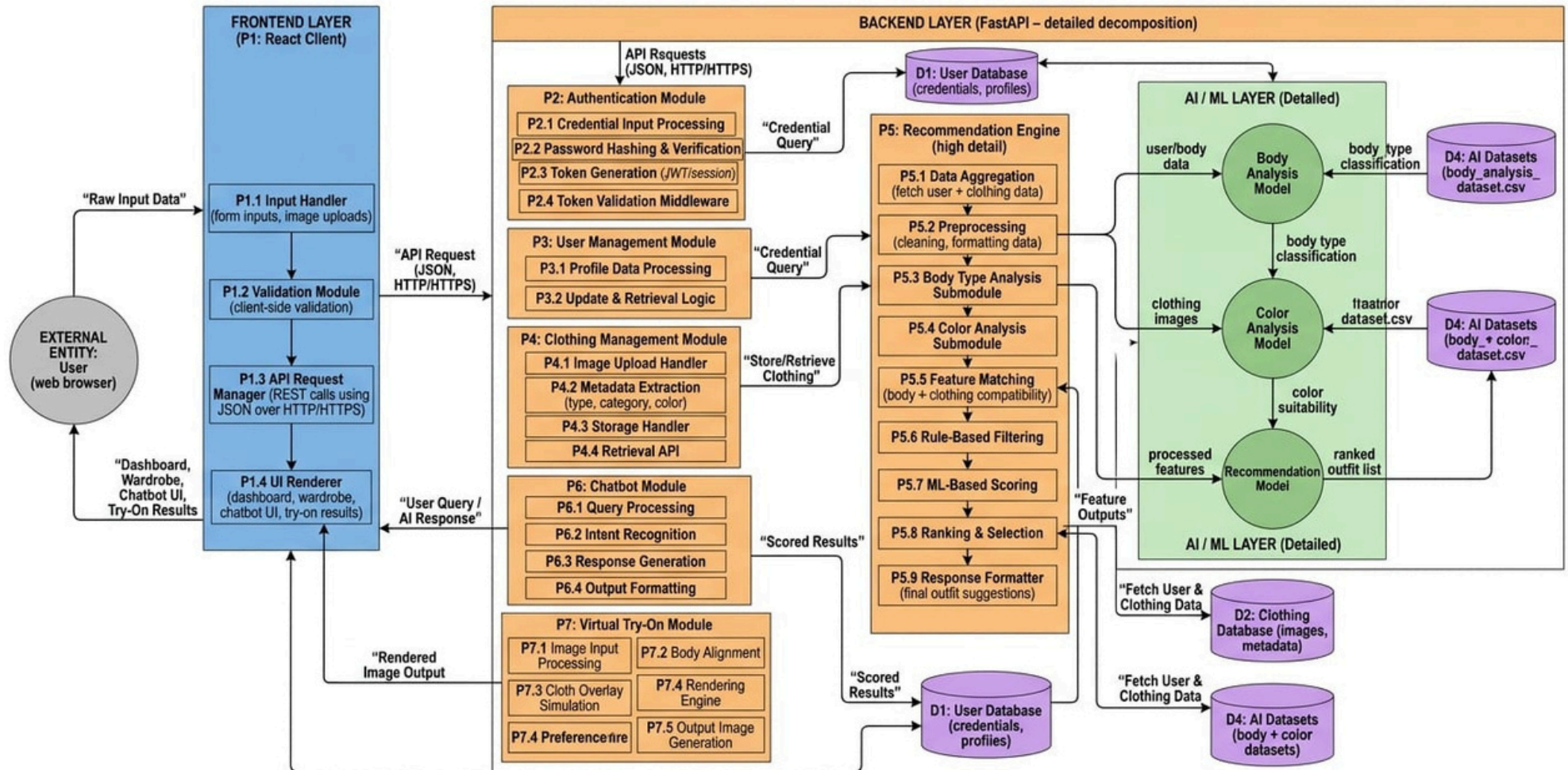
Full-stack AI-based web application (DFD Level 2)



DATA FLOW DIAGRAM

Data Flow Diagram (DFD Level 3) for AI-powered full-stack web application

Clueless – AI Fashion Stylist



IMPLEMENTATION

Frontend Implementation

- Built using React.js with TypeScript and Vite
- React Router for navigation across modules
- Tailwind CSS and Radix UI for responsive UI
- Protected routes for authenticated access

Backend Implementation

- Developed using FastAPI (Python)
- Modular APIs for authentication, users, clothing, chatbot, and try-on
- Handles validation, business logic, and AI integration

Database Implementation

- PostgreSQL with SQLAlchemy ORM
- Alembic for database migrations
- Stores user data, wardrobe, wishlist, and outfits

IMPLEMENTATION

Authentication and Security

- JWT-based authentication
- Password hashing using bcrypt (PassLib)
- Token-based API protection and CORS enabled

Clothing Management

- Upload and manage clothing images
- Images stored in compressed base64 format
- Metadata: category, color, style, occasion, mood, weather
- Supports favorites and wishlist

AI Classification & Datasets

- Integrated Google Gemini API for clothing analysis
- Extracts type, color palette, style, and occasion suitability
- Uses CSV-based datasets for body shape and color analysis
- Maps user appearance to personalized styling recommendations

IMPLEMENTATION

AI Stylist

- Chatbot powered by Gemini API
- Uses user profile and wardrobe data
- Provides personalized styling advice and updates user preferences

Virtual Try-On

- Integrated IDM-VTON via Hugging Face
- Processes user and garment images
- Generates realistic try-on outputs

Outfit Recommendation & E-Commerce Integration

- Generates outfits based on wardrobe and AI logic
- Filters by occasion, mood, weather, and style
- Integrated SerpAPI for Shopping results
- Enables product discovery and purchase options
- Allows saving outfits

IMPLEMENTATION

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RESULTS

clueless

Sign in

Get Started

AI - Powered Personal Styling

Your closet, *reimagined.*

Clueless is your AI stylist that learns your taste, curates outfits, and transforms the way you dress - effortlessly.

Start Styling →



"This outfit is so you!"
- Your AI Stylist

RESULTS

Your personal AI stylist is
one step away.

Join Clueless and never have a "clueless" moment again.

clueless
Create your account

Start your style journey today.

Email

you@example.com

Username

Choose a username

Password

Create Account

Already have an account? [Sign in](#)

clueless
Welcome back

Sign in to your account to continue styling.

Email or Username

sritejaswini15@gmail.com

Password

[Forgot password?](#)

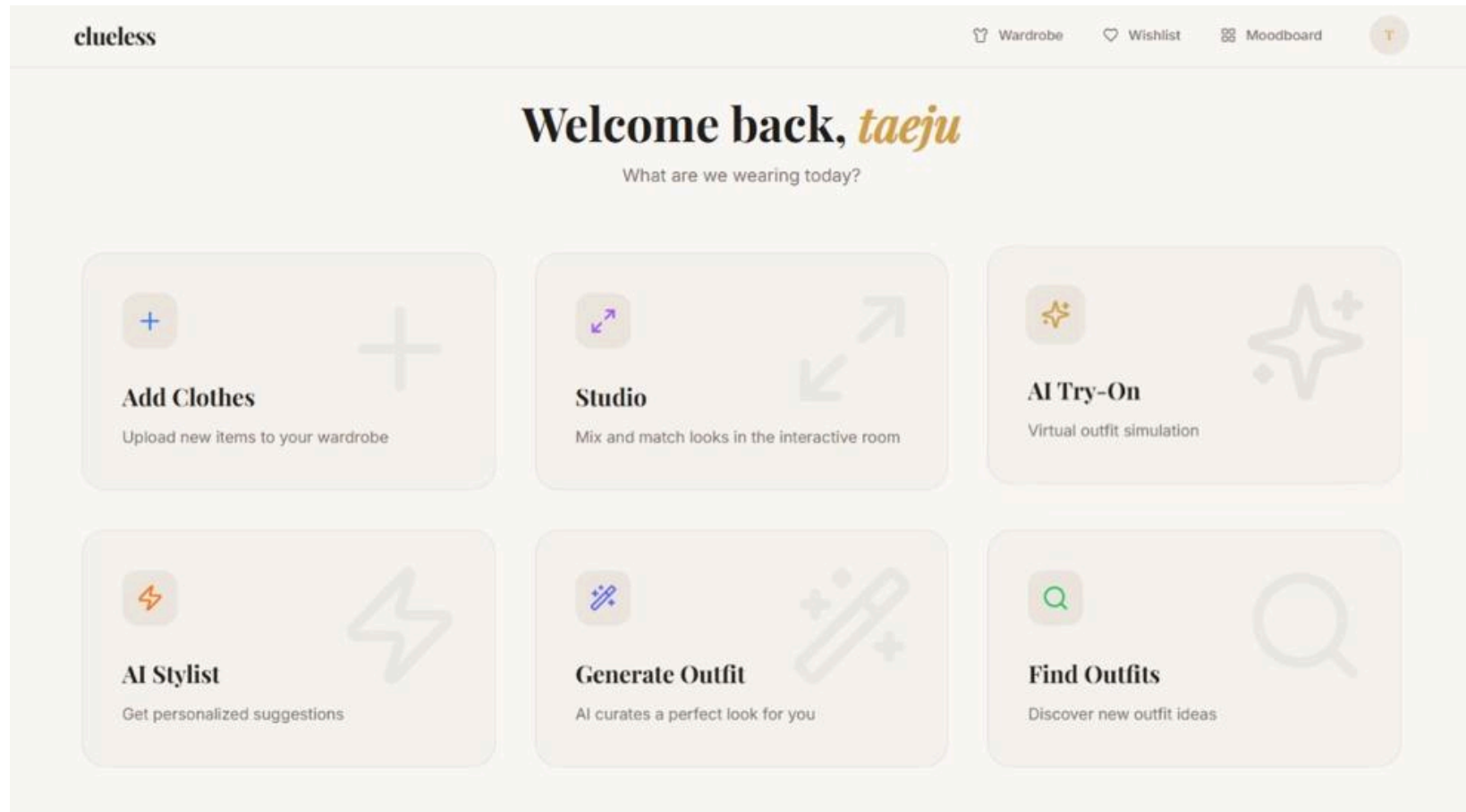
Sign In

Don't have an account? [Sign up](#)

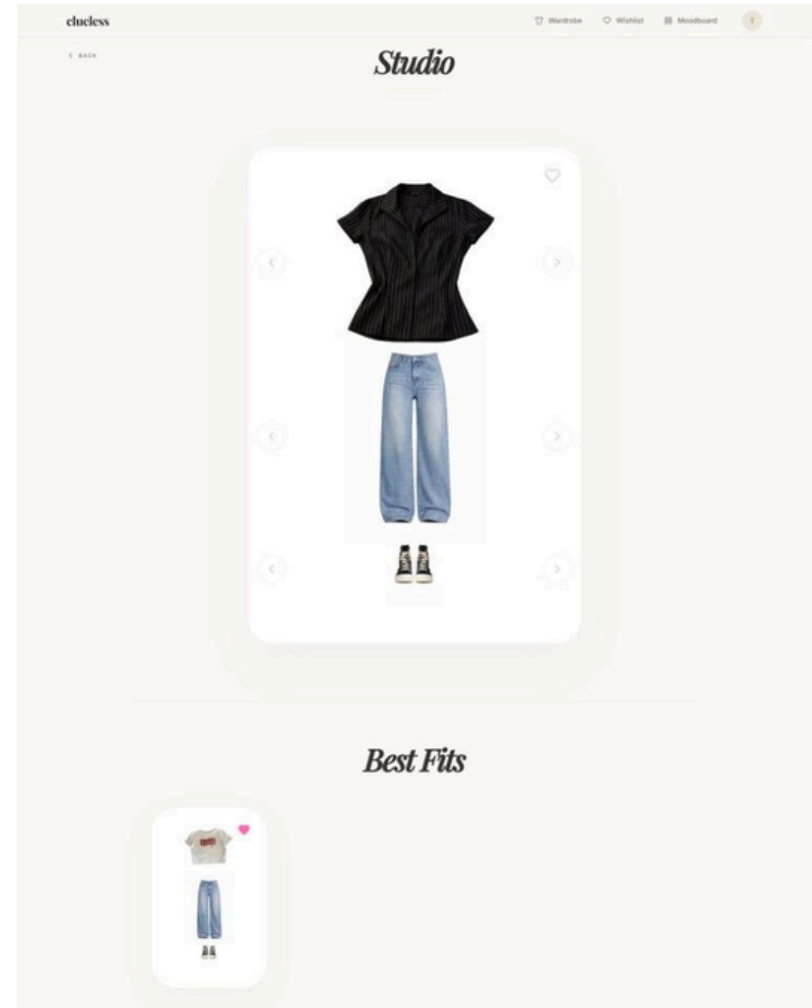
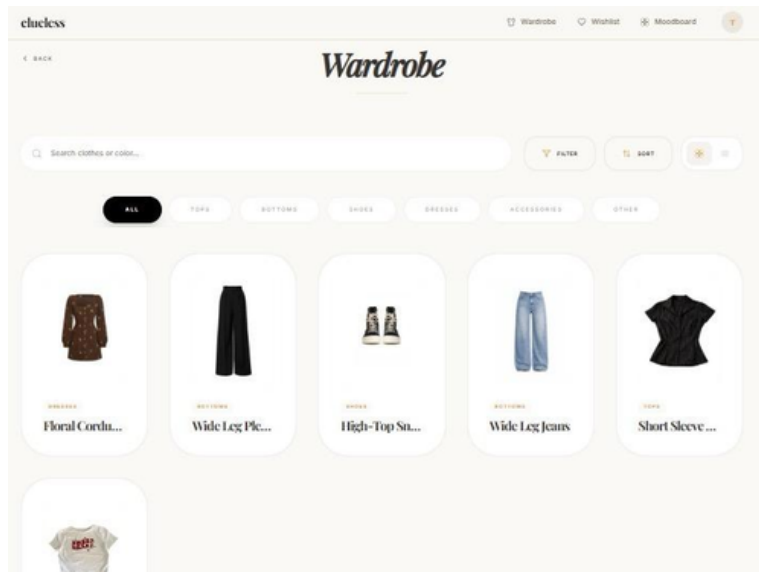
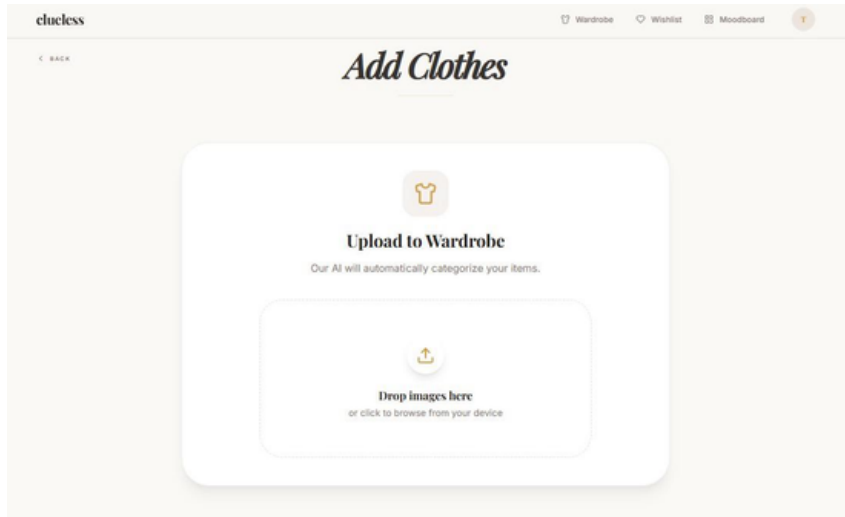
"As if I'd leave the house
without checking *Clueless*
first."

- Every stylish person, probably

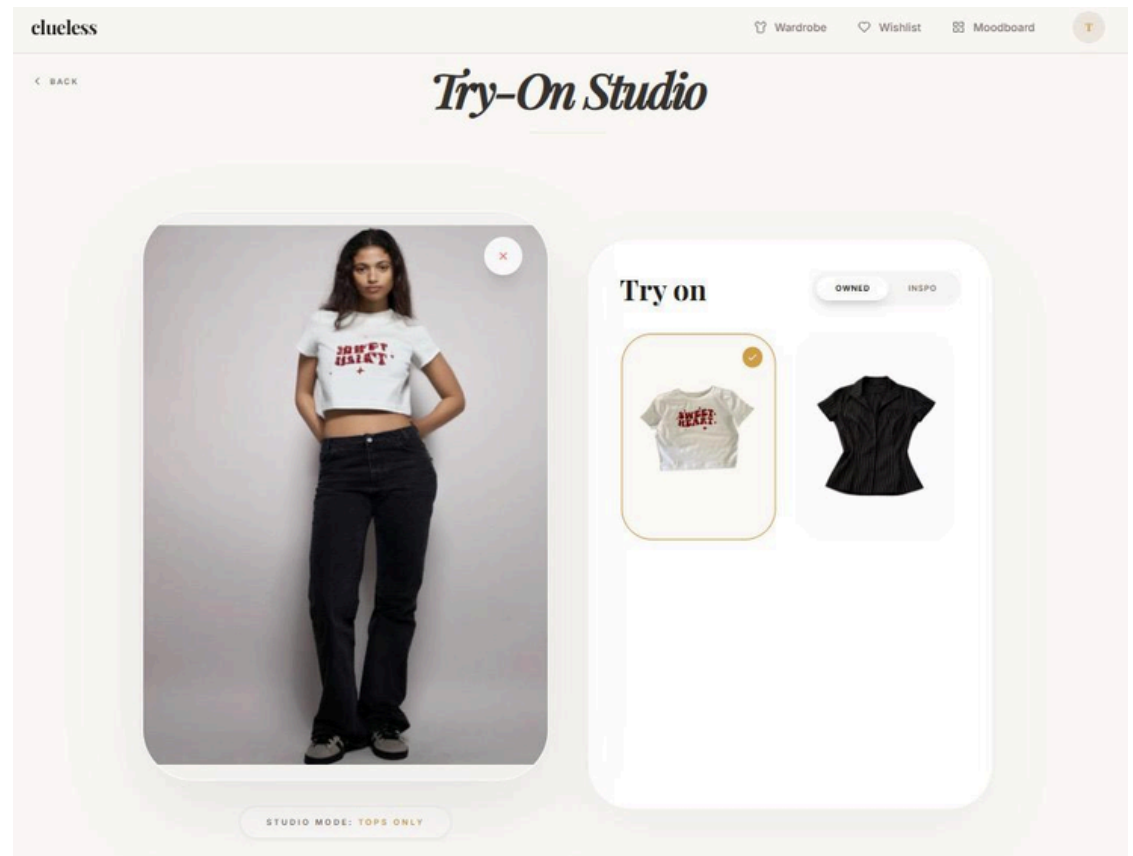
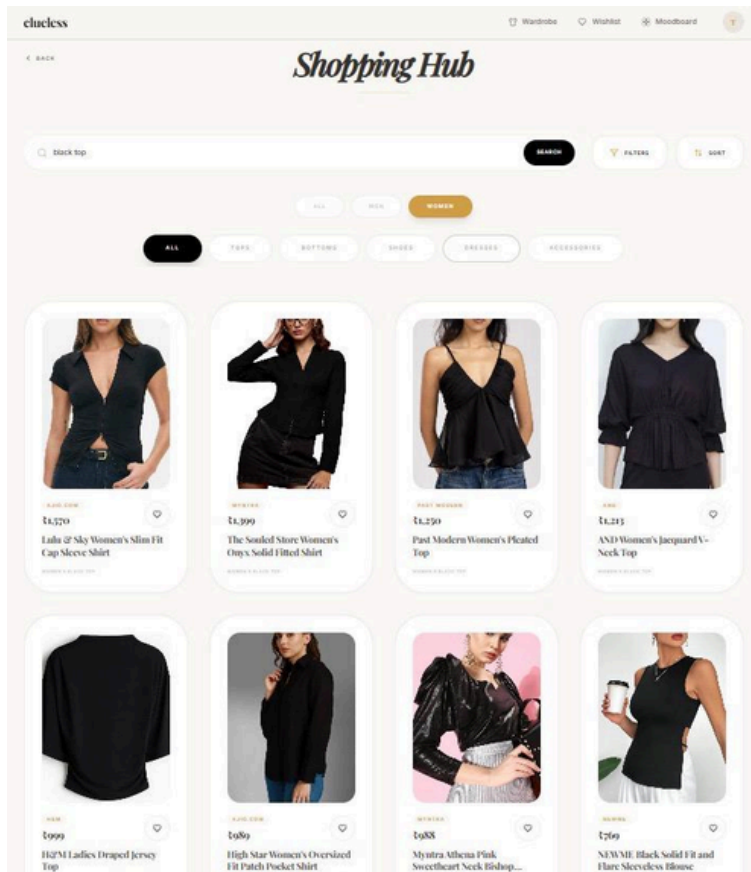
RESULTS



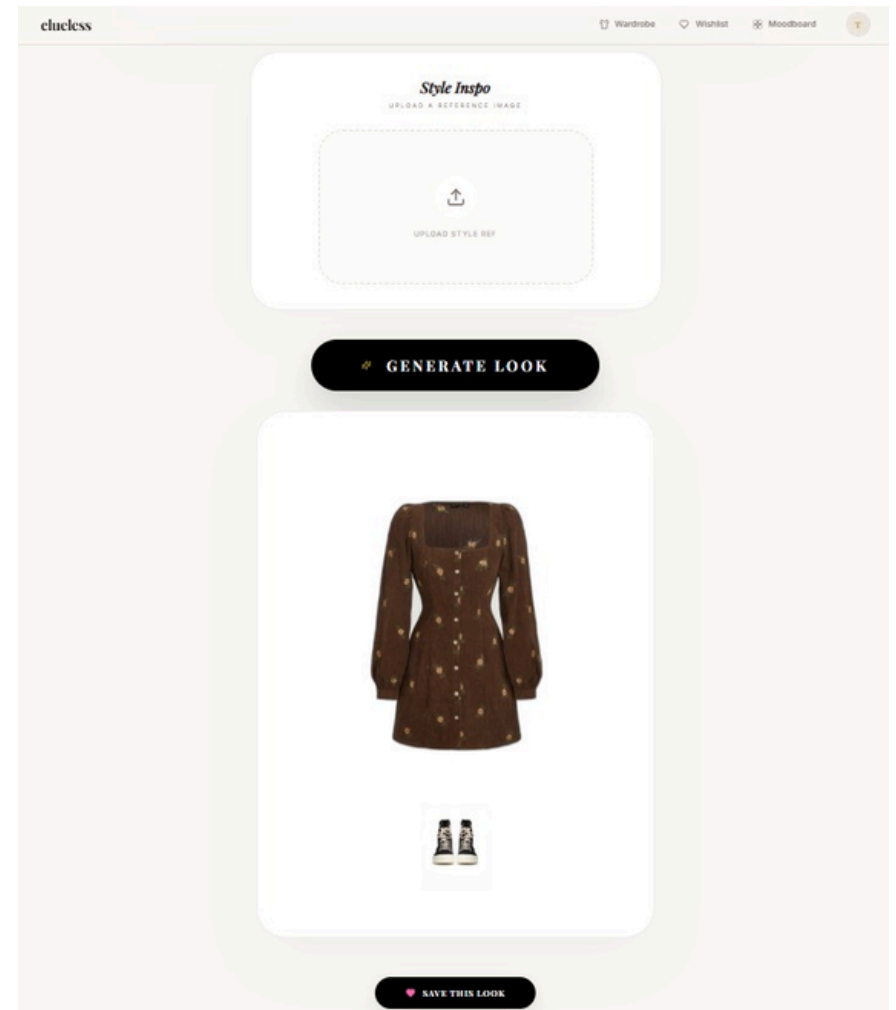
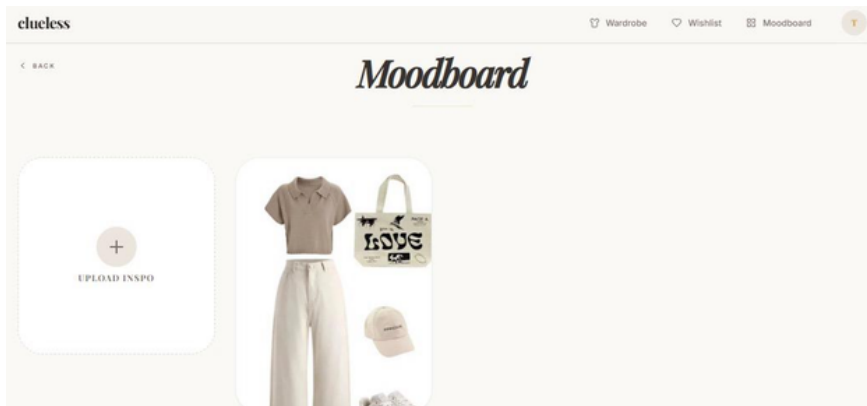
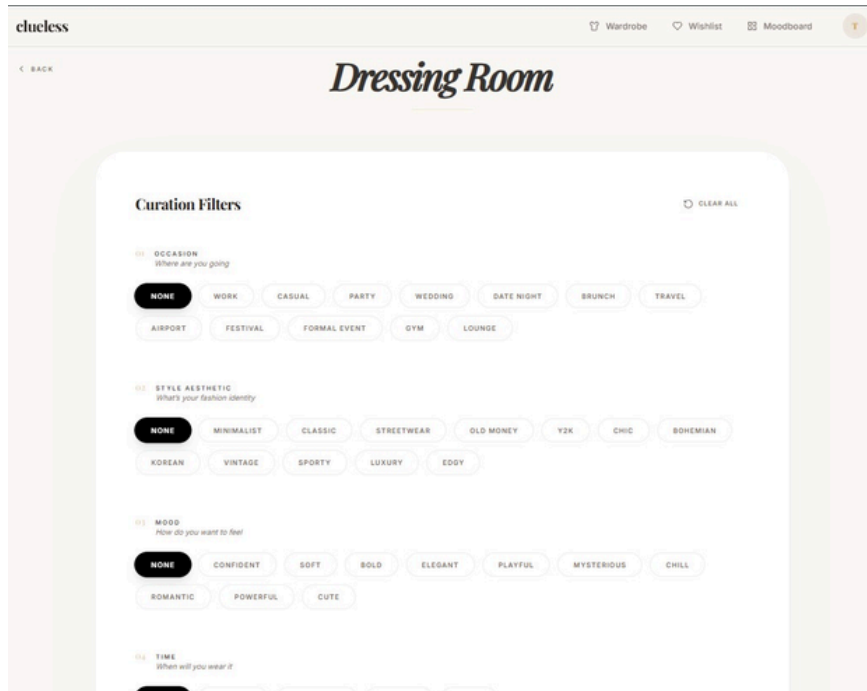
RESULTS



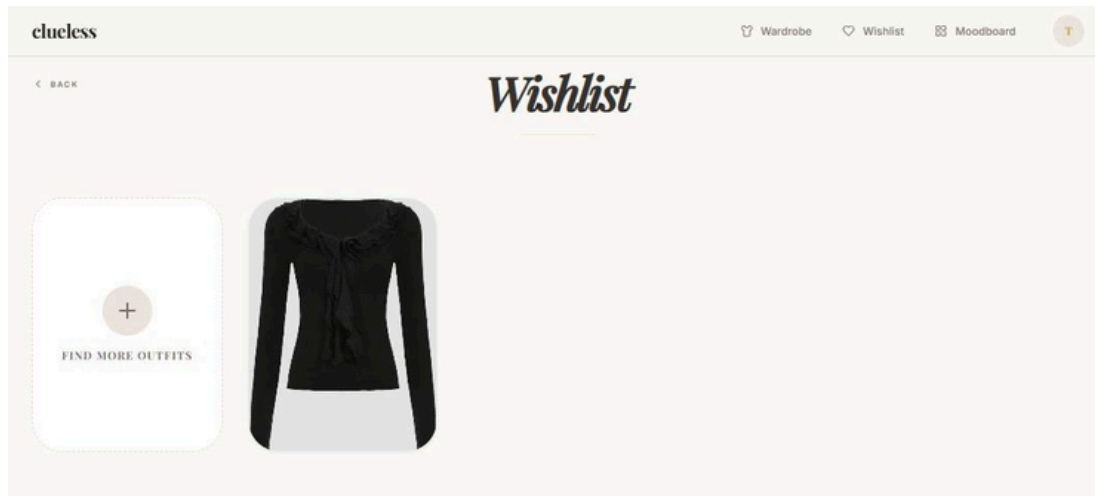
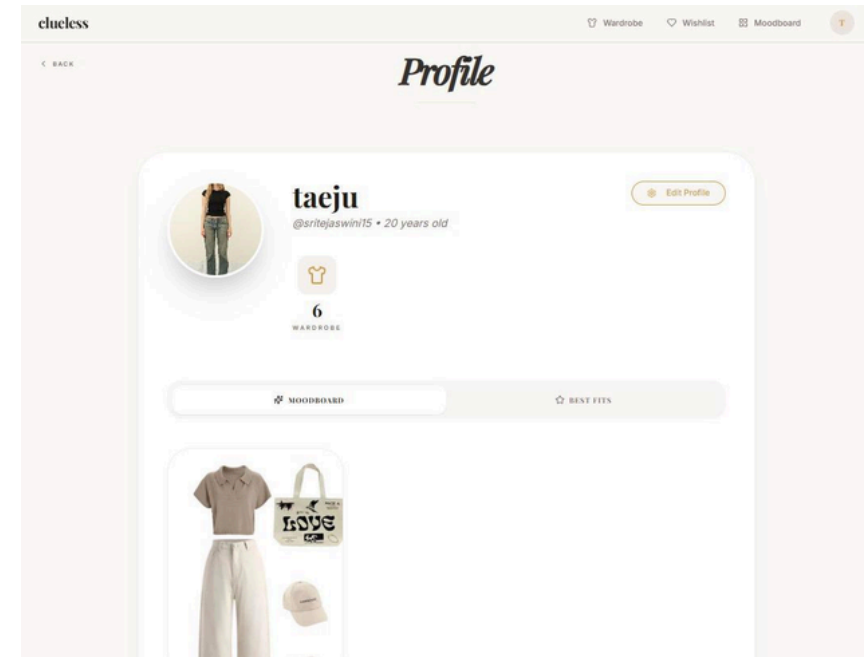
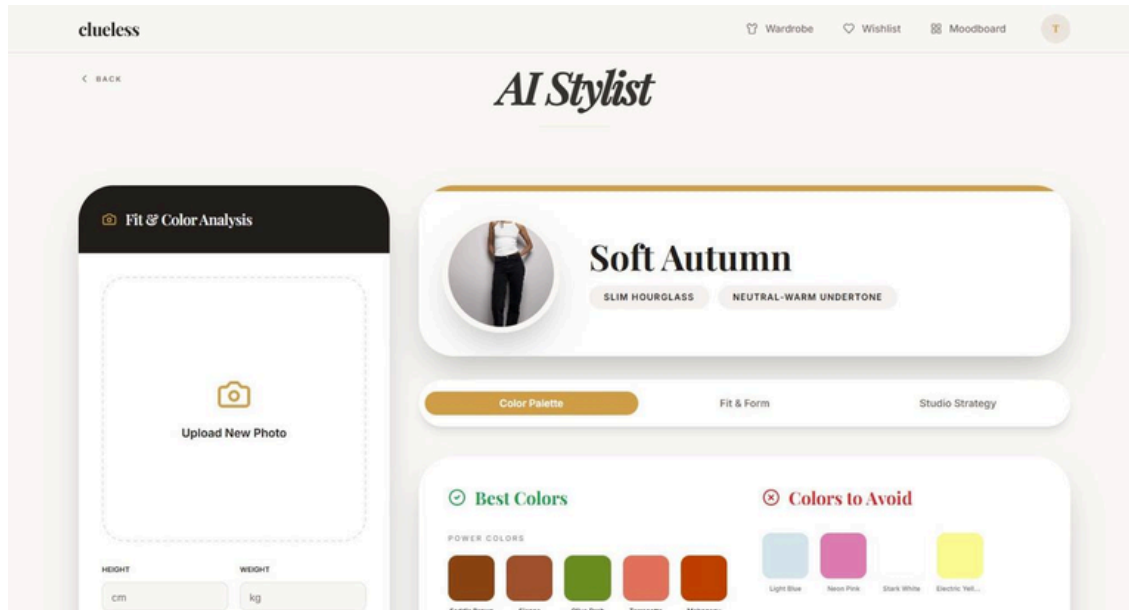
RESULTS



RESULTS



RESULTS



TESTING & VALIDATION

Test Case ID	Test Description	Test Steps	Expected Results	Actual Results	Pass/Fail
TC01	User Registration	Enter valid user details and submit sign-up form	User account should be created successfully	User account created and redirected to login/dashboard	Pass
TC02	User Login	Enter valid email and password and click login	User should be authenticated and redirected to dashboard	Login successful and dashboard opened	Pass
TC03	Invalid Login	Enter incorrect email or password	System should display error message and deny access	Error message displayed, login blocked	Pass
TC04	Protected Route Access	Try to open dashboard without logging in	System should prevent access and redirect to login page	Unauthorized access blocked and redirected	Pass
TC05	Profile Update	Edit profile details and save changes	Updated profile information should be stored successfully	Profile updated and displayed correctly	Pass
TC06	Clothing Upload	Upload clothing image with required details	Clothing item should be stored in wardrobe	Clothing item uploaded and visible in wardrobe	Pass

TESTING & VALIDATION

Test Case ID	Test Description	Test Steps	Expected Results	Actual Results	Pass/Fail
TC07	AI Clothing Classification	Upload a clothing image for analysis	System should classify clothing attributes such as category, color, and style	Clothing metadata generated correctly	Pass
TC08	Wardrobe Display	Open wardrobe page after uploading items	All uploaded clothing items should be displayed properly	Wardrobe items displayed correctly	Pass
TC09	Wishlist Management	Add an item to wishlist	Selected item should be stored in wishlist	Item added successfully to wishlist	Pass
TC10	Outfit Generation	Request AI-generated outfit suggestion	System should generate outfit combinations based on wardrobe and profile	Outfit suggestions generated successfully	Pass
TC11	AI Stylist Chatbot	Ask styling-related question in chatbot	Chatbot should return personalized fashion advice	Relevant response generated based on user data	Pass
TC12	Virtual Try-On	Upload person image and garment image, then start try-on	System should generate try-on output image	Try-on image generated successfully	Pass

TESTING & VALIDATION

Test Case ID	Test Description	Test Steps	Expected Results	Actual Results	Pass/Fail
TC13	Moodboard Feature	Add inspiration image to moodboard	Image should be stored and displayed in moodboard	Moodboard image saved and shown correctly	Pass
TC14	Logout Functionality	Click logout button	User session should end and user should return to home/login page	Logout successful and redirected	Pass
TC15	Database Storage Validation	Perform user, wardrobe, and wishlist operations	Data should be stored and retrieved correctly from database	Data stored and fetched correctly	Pass

CONCLUSION

- Developed an AI-powered personal styling and wardrobe management platform
- Addresses wardrobe organization, outfit selection, and personalization challenges
- Integrates wardrobe management, AI analysis, recommendations, chatbot, moodboard, wishlist, and virtual try-on
- Built using React, TypeScript, FastAPI, PostgreSQL, Google Gemini, Hugging Face, and IDM-VTON
- Demonstrates practical use of AI, computer vision, NLP, and recommendation systems
- Provides reliable performance and improves user decision-making

FUTURE WORK

- Improve outfit recommendation accuracy using advanced machine learning techniques
- Expand virtual try-on to support more clothing types and full outfits
- Integrate weather and location-based outfit suggestions
- Develop a mobile application for better accessibility
- Connect with e-commerce platforms for product purchasing
- Add voice and multilingual support to the AI stylist
- Introduce social sharing and fashion inspiration features
- Include analytics for tracking wardrobe usage
- Extend the system into a complete smart fashion assistant

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